Application of Gantt Charts in the Educational Process

Andrea Bednjanec,
School of Electrical Engineering (Elektrotehnička škola), Zagreb, Croatia
andrea.bednjanec@gmail.com

Martina Filipović Tretinjak,
School of Electrical Engineering (Elektrotehnička škola), Zagreb, Croatia
martinafilipovic.tretinjak@gmail.com

Abstract - Gantt charts, a project management tool for planning and scheduling projects, can be used in educational processes. With the increasing duties, individuals are responsible to effectively manage time and complete tasks. An advantage of Gantt charts is their graphical overview. They visualize beginning dates, ending dates and the important duration parameters helping to track various projects over specific time periods. In the educational process there are many tasks which need to be planned such as school activities, school curriculum, competitions, school projects, intern´s activities, internship worksheets, teaching process, introducing curriculum fields, allocating time of the certain curriculum fields and subjects, defining efficient class strategies showing how many time will be spent doing various activities, guidance for lesson development, self-evaluation, etc.. Gantt chart's graphical representation is also very useful in the field of the teacher preparation. There are three components important in teacher preparation: teacher knowledge of the subject to be taught, knowledge and skill in how to teach and determining the time educational content will be introduced. Experience from the School of Electrical Engineering in Zagreb shows that use Gantt chart's in self-evaluation, intern´s activities planning, lesson planning and teacher preparing make teaching more effective.

I. Introduction

A table is the most common planning tool in the educational process. But they do not give a visual overview of all planned activities or events. Using them, it is impossible view the time of events that partially or completely disrupt planned tasks. In addition, tables and text files do not provide a visual connection ability nor the sequence of events which the pieces of the plan.

In the School of Electrical Engineering in Zagreb some teachers make researches on usage Gantt charts in the teacher preparation and planning the internship activities and teaching process. To facilitate the planning of the teaching process and ongoing monitoring and self-evaluation of our work and achieve higher levels of educational outcomes, we decided to improve planning some segments of our educational work with Gantt charts The result of this work is better preparation of teachers and their ability to easier organize and maintain the educational process. That results in greater satisfaction in everyday teacher´s work and achievements greater competences of students and their satisfaction with the educational process.

Gantt charts visualize each distinct activity as a block of time. Their main advantage is graphical overview. Each block of time is labeled with a title to show the task and the amount of time that the block represents. Gantt Charts the time each activity is expected to take and the order in which activities should occur. Each plan needs to be detailed with large tasks broken down into smaller subtask what makes easier to keep track of how it is getting further.

They are used for planning the time overall project would take and where the pressure points in the project will occur. Gantt charts are helpful tool to see immediately what should have been achieved at any point in time. They are also help in troubleshooting of potential delays seeing how remedial action may bring the project back on course.

It was found that this way of preparation and planning segments of teaching process is more effective, and that usage of Gantt charts can be involved in more school activities, school curriculum, competitions, etc.

We explored on the internet if other teachers in the world have experience with the use of Gantt charts, but we haven´t found a lot of documents. We have found only a few examples. Staff of the Francis Marion University, South Carolina, USA, has made Gantt chart to provide a general timeline of activities designed to help their University to earn re-accreditation. Institute of Technical Education Singapore use Gantt charts for Time Management Schedule in Completion of Supervised Field Teaching of participants who acquire the Pedagogical Certificate in Technical Education. University of Canberra has organized work in groups defining and setting time frames using Gantt charts. Board of Regents of the University System of Georgia (developed by its GLISI, Georgia's Leadership Institute for School Improvement) in A Performance-based Learning Module for Georgia's Educational Leaders has recommended Gant charts as a project management tool. The Trafalgar School at Downton, UK, uses Gantt charts to teach students how to learn effectively and to take responsibility for their own time management.

Gantt charts can be created using the many free tools from the Internet, and can be used templates in MS Office (MS Word, MS Excel). Gantt chart can even be made in pencil on paper, without the use of computers. Their advantage is a deeper analysis of the problems we are discussing and raising awareness of key points where problems may arise. Shift plan is easily resolved in electronic form, what improves prospects for its final execution. We usually use KS Project Planner 2011, a freeware tool for Windows XP to Windows 8, 32-bit and 64-bit. However, we are aware that at the moment this kind of planning is not suitable for teachers who are not well in computer science.

Application of Gantt charts in the educational process is discussed is the SWOT analysis given in Table 1.

Based on the SWOT analysis can be concluded that for a wider use of Gantt charts in the collective should be a critical mass of people with ICT skills and the will to engage themselves.

TABLE I. SWOT ANALYSIS OF THE GANTT CHARTS APPLICATION IN THE EDUCATIONAL PROCESS

	Strengths	Weaknesses
Internal	1. Graphical overview used in planning educational processes. 2. Simple way to monitor and evaluate progress through schedule status. 3. Easy way to manage deviations from the curriculum. 4. Easy to detect disadvantages inside the curriculum. 5. No software implementation costs.	1. Takes time to create Gantt chart. 2. Changes to the schedule require a redrawing of the chart. 3. Curriculum structure must be completed before the chart can be drawn.
	Opportunities	**Threats**
External	1. Good reputation as an innovative educational institution. 2. Creating an innovative working climate. 3. Provides efficient time management. 4. Enables effective distribution of activities.	1. There is no positive attitude in the school towards change. 2. Some staff members are reluctant to learn necessary IT skills. 3. Gantt chart should not deviate from the specified plans and programs.

II. EXPERIENCES WITH USE OF GANNT CHARTS IN PLANNING

Our research is based on experience in the planning internship activities in planning during the educational process, planning and scheduling of lessons, educational outcomes and student´s projects, and thus preparing teachers and classes constantly just evaluating our teaching.

A. Internship activities planning

One of the most important components in preparing new teachers is a successful internship experience. During internship program intern is involved in many tasks which need to be organized through different activities such as: participating in faculty and school committee meetings, attending professional development seminars, collaboration activities between mentor and intern, keeping an internship diary and taking part in various school and extracurricular activities.

Most activities are carried out simultaneously. In order to increase the efficiency of the internship and to stimulate the motivation of the intern, internship program needs to be carefully planned. Duration of each activity varies across the year and depends on the correlation between mentor and intern's curriculum.

The main purpose of Gantt chart is to notice and if it's possible to avoid overlapping of intern's tasks. It provides a simple way to monitor and evaluate intern's progress through activities that have been completed. Through Gantt chart intern can also learn self-evaluation which enables him to achieve high quality educational outcomes and become an active participant in his professional improvement.

Chart shown in Figure 1 is based on a common school calendar, in which there are two semesters. At the beginning of the internship, intern observes mentors' teaching to achieve basic competencies. Afterwards, mentor observes interns' teaching and in meetings gives post-observation feedback for the interns' lessons.

During entire internship, intern is taking notes what allow him to record and apply mentors' teaching methods.

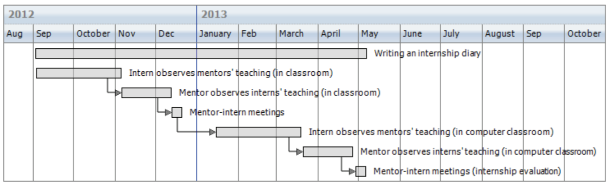

Figure 1. Internship activities planning

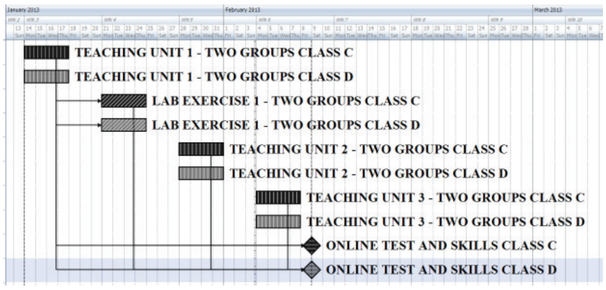

Figure 3. Gantt chart schedule planning lessons and activities in Computer Networks

B. *Planning and scheduling of lessons*

Over the last 5 years Gantt diagrams are applied for schedule planning of lessons and lab exercises in the Electrical Measurements and in the Computer Networks. The goal of planning is to enable students to achieve theoretical learning outcomes required for the acquisition of competencies by performing laboratory exercises on time.

Figure 2 shows a Gantt chart schedule planning lessons for one class. Connection of instruments to measure current, voltage, resistance, frequency in the circuit is labeled as TEACHING UNIT 1, Connection of the wattmeter in a circuit - direct measurement of active power is labeled as TEACHING UNIT 2, Repeat of wattmeter is labeled as TEACHING UNIT 3, Semi-direct measurement of active power is labeled as TEACHING UNIT 3, and Measurement of active power by wattmeter in direct connection is labeled as LABORATORY EXERCISE. Scheduling depends on students´ timetable and is different for each class.

Figure 3 shows a Gantt chart schedule planning lessons and activities for two of third classes attending the same program (Computer Technicians). Each class is divided in two groups. Most of students from those two classes attends Cisco Networking Academy classes to and prepare from Certificates. They have Cisco lessons Saturdays, every two weeks. For that classes they have to be good prepared, reaching educational achievements on the Cisco Academy CCNA level. Gantt charts are helpful in scheduling teaching units, computer exercises, exercises on equipment, online tests, etc. Description of the router is labeled as TEACHING UNIT 1, Basic router configuration lab exercise is labeled as LAB EXERCISE 1, Construction of routing tables is labeled as TEACHING UNIT 2, Routing and switching function tables is labeled as TEACHING UNIT 3. Static routing is labeled as TEACHING UNIT 4. Cisco Networking Academy activities are labeled as ONLINE TESTS AND SKILLS.

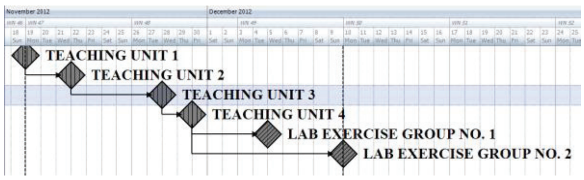

Figure 2. Electrical Measurements schedule planning

In last 5 years research is made on 120 randomly selected students from third class attending the both, Computer Networks and Cisco Academy, CCNA program. Their lessons and educational achievements were planned using Gantt diagrams. Figure 4 shows results of researches.

C. *Students´project activities planning*

In the last two years students of the fourth class are involved in projects based on collaboration among Internet Technologies competences (Computer Technicians) and Politics and the Economy educational achievements Technicians for Electrical Machines with Applied Computer Science). During the school year they make three projects. Students involved with Politics and the Economy educational achievements order from students who are involved with Internet Technology skills a poster of imaginary political party, a promo-video for that party and a web site of an imaginary firm. They order the same thing in three classes and in each project choose one work as the best what meets their demands. Students are very engaged in those projects and work on them very likely.

Figure 5 shows Gantt chart of student´s project

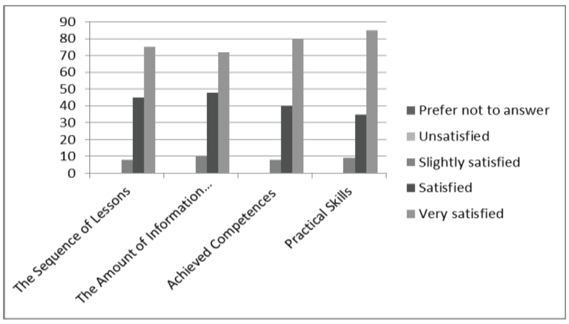

Figure 4. Research on satisfaction of students about teaching process in Computer Networks

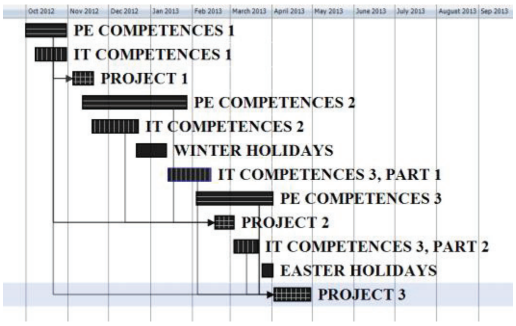

Figure 5. Gantt chart shedule planning student´s projects

activities schedule. PROJECT 1 is made after Computer Technicians achieve competences in editing of pictures (IT COMPETENCES 1) and Technicians for Electrical Machines with Applied Computer Science get competences to order appropriate pictures, text and color combination for an imaginary political party (PE COMPETENCES 1). After that Computer Technicians achieve competences in creating a video from pictures, and Technicians for Electrical Machines with Applied Computer Science get competences (PE COMPETENCES 2) to order appropriate promo-video for their imaginary parties. Computer Technicians in PROJECT 2 make videos (based on IT COMPETENCES 2) and must obey their demands doesn´t matter if their views are different from political party they work for. PROJECT 3 is made when Computer Technicians achieve competences in creating web site from a template (IT COMPETENCES 3) and Technicians for Electrical Machines with Applied Computer Science learn what is important for promotion of the business (PE COMPETENCES 3).

III. Possibilities of Gantt Charts Application in the School Management

In the school management there are many possibilities of Gantt charts application. The aim of that tool is to improve school curriculum planning, managing different activities and time. Two common examples are possibility of schedule teaching units based on correlations among the subjects, and planning of school activities as a part of the school curriculum. Involving of Gantt charts in the school management planning can be complicated in some schools because it demands a big engagement of different school stuff.

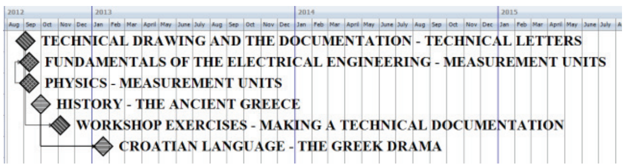

Figure 6. Gantt chart correlation among subjects schedule

A. *Gantt charts based on correlations among the subjects*

In the School of Electrical Engineering in Zagreb, for example, in the first class there are a lot of teaching units that correlate. Similar situation is in the upper classes, too. When a teaching stuff work as a team planning together the levels of educational achievements, results would be saving the time and reaching a higher levels student´s competences.

When is needed to set the beginning of an action or event with the completion of the previous one, is practically to display it using Milestone chart. Diagram shows only the events that must be brought into line. Figure 6 shows a Milestone Gantt chart based on beginning of several correlations in the first months of the school year. Those correlations are among: History (The Ancient Greece) and the Croatian Language (The Greek Drama), Fundamentals of the Electrical Engineering (Measurement Units) and Physics (Measurement Units), Technical Drawing and Documentation (Technical Letters) and Workshop Exercises (Making of Technical Documentation).

B. *Gantt charts based on planning the school curriculum activities*

Gantt chart tools enable permanent notes about the state of the each phase of the project. If an activity lasts a long time, it is possible to calculate what part of the activity is finished.

It is shown in the Figure 7. School curriculum activity IPA results implementation in the classroom activities – Career Guidance to Students lasts from the September to the June. It includes five Workshops and the one sub-activity for students, called Career Based Corner. The picture shows the situation in January 2013. Two of workshops are the part of that activity: Workshop Writing a CV, the application, simulation of talks with the employer is already finished, and workshop Self-employment will begin. Different labeling graphs can be distinguished actions that have been completed.

Figure 7. Gantt chart of the school curriculum activity: IPA results implementation in the classroom activities – Carrier Guidance to Students

IV. CONCLUSION

The Gantt charts application in the educational process improves planning and makes a teaching easier. This way of preparation and planning segments of teaching process is more effective. Besides a planning of teaching process and lessons scheduling usage of Gantt charts can be involved in more school activities, school curriculum, competitions, etc. Selection of the Gantt charts software depends on application of the chart and on user´s ICT skills.

REFERENCES

[1] http://www.gantt.com/

[2] Elektrotehnička škola, Zagreb, Konavoska 2: Školski kurikulum 2012/2013

[3] http://www.docstoc.com/docs/51927218/Gantt-Chart-for-Planning-Accreditation-Activities

[4] http://www.docstoc.com/docs/135594311/Academic-Year-Simple-Gantt

[5] http://edt.ite.edu.sg/eMentor/04_htm/sftmaterials_04/TimeMgt_04.doc

[6] http://edt.ite.edu.sg/eMentor/07_htm/sftmaterials_07/TimeMgt_07.doc

[7] http://www.canberra.edu.au/studyskills/learning/groups

[8] http://www.galeaders.org/sites/default/files/pbm/Leading%20a%20Team%20to%20Develop%20Project%20Management%20Tools.pdf

[9] http://www.trafalgarschool.com/

[10] http://www.kroll-software.de/products/projectplanner_e.asp

[11] http://www.businessdictionary.com/definition/Gantt-chart.html

[12] Elektrotehnička škola, Zagreb, Konavoska 2: Godišnji plan I program rada 2012/2013

[13] http://edt.ite.edu.sg/ActionResearch/index.htm

[14] http://www.easyfreeware.com/freeware/gantt_chart.html

[15] http://www. cisco.com